

Rigid-Body Rotations

1 Angular Momentum, Torque, and Energy for Rotation about a Fixed Axis

For rotation about a fixed axis (the z -axis),

$$L_z = I\omega_z, \quad \vec{N} = \vec{P} \times \vec{F}, \quad T = \frac{I}{2}\omega_z^2 = \frac{\omega_z L_z}{2}, \quad (1)$$

where $\vec{P}$ is the position vector of the point of application of the force $\vec{F}$. The rate of change of angular momentum equals the external torque,

$$\frac{d\vec{L}}{dt} = \vec{N}^{(e)}, \quad (2)$$

evaluated either in an inertial frame, or in the center-of-mass (CM) frame (origin at the CM, axes parallel to the inertial axes).

1.1 Rolling without Slipping

For a wheel of radius R rolling on a horizontal surface (Fig. 1), the rolling-without-slipping condition is

$$v_{CMx} + \omega_z R = 0, \quad (\omega_z < 0 \text{ for clockwise rotation}). \quad (3)$$

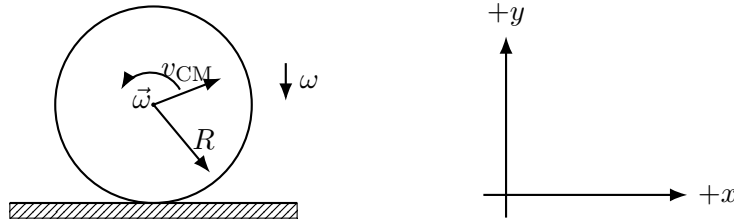


Figure 1: Rolling without slipping: a wheel of radius R with angular velocity $\vec{\omega}$ (clockwise for the sense drawn) and center-of-mass velocity v_{CM} , resting on a fixed surface, together with the sign convention for the x - y plane.

1.2 The Parallel-Axis Theorem (Elementary Form)

For a rigid body rotating about a fixed axis a distance a from the center of mass (Fig. 2),

$$I = \bar{I} + Ma^2, \quad (4)$$

where $\bar{I}$ is the moment of inertia about the parallel axis through the center of mass and a is the distance between the axis of rotation and the center of mass.

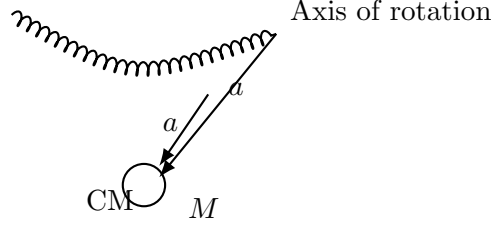


Figure 2: The parallel-axis theorem: the axis of rotation lies a distance a from the center of mass of an otherwise arbitrarily shaped rigid body.

1.3 Separating the Center-of-Mass Motion

For a system of particles with masses m_j and positions $\vec{r}_j$, define the total mass, center-of-mass position, and positions relative to the center of mass:

$$M = \sum_j m_j, \quad \vec{R} = \frac{\sum_j m_j \vec{r}_j}{M}, \quad \vec{r}'_j \equiv \vec{r}_j - \vec{R} \quad (\text{position relative to CM}). \quad (5)$$

The total angular momentum and kinetic energy separate as

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}'_j \times m_j \dot{\vec{r}}'_j, \quad T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} \sum_j m_j |\dot{\vec{r}}'_j|^2. \quad (6)$$

For a rigid body rotating with angular velocity ω_z about the CM,

$$L_z = \left(\vec{R} \times M \dot{\vec{R}} \right)_z + I \omega_z, \quad T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{I}{2} \omega_z^2. \quad (7)$$

Figure 3 shows the geometry: position vectors $\vec{r}_i$ from a fixed origin to each particle, the center-of-mass vector $\vec{R}$, and the relative vectors $\vec{r}'_i = \vec{r}_i - \vec{R}$.

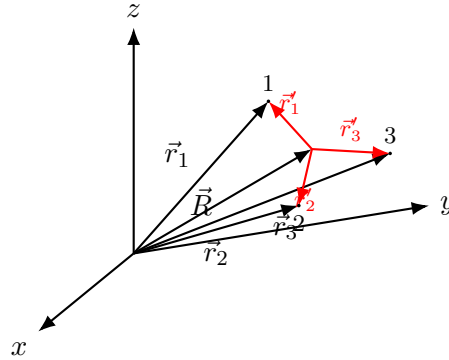


Figure 3: Position vectors $\vec{r}_i$ of individual particles measured from a fixed origin, the center-of-mass vector $\vec{R}$, and the vectors $\vec{r}'_i = \vec{r}_i - \vec{R}$ relative to the center of mass.

Proof

Changing to the CM system, write $\vec{r}_i = \vec{R} + \vec{r}'_i$ and $\dot{\vec{r}}_i = \dot{\vec{R}} + \dot{\vec{r}}'_i$. Then

$$\vec{L} = \sum_{i=1}^N \vec{r}_i \times m_i \dot{\vec{r}}_i = \sum_{i=1}^N (\vec{R} + \vec{r}'_i) \times m_i (\dot{\vec{R}} + \dot{\vec{r}}'_i) = \sum_i \vec{R} \times m_i \dot{\vec{R}} + \sum_i \vec{r}'_i \times m_i \dot{\vec{r}}'_i + \text{cross-terms}. \quad (8)$$

The cross-terms vanish identically, using $M\vec{R} = \sum_i m_i \vec{r}_i$ and its time derivative (Newton's second law for the particle system, $M\ddot{\vec{R}} = \sum_i \vec{F}_i^{(e)}$, is not yet needed here):

$$\sum_i \vec{r}_i' \times m_i \dot{\vec{R}} = \sum_i m_i (\vec{r}_i - \vec{R}) \times \dot{\vec{R}} = \left[\underbrace{\sum_i m_i \vec{r}_i}_{M\vec{R}} - \left(\sum_i m_i \right) \vec{R} \right] \times \dot{\vec{R}} = [M\vec{R} - M\vec{R}] \times \dot{\vec{R}} = \vec{0}, \quad (9)$$

$$\sum_i \vec{R} \times m_i \dot{\vec{r}}_i = \vec{R} \times \sum_i m_i (\dot{\vec{r}}_i - \dot{\vec{R}}) = \vec{R} \times \left[\underbrace{\sum_i m_i \dot{\vec{r}}_i}_{M\dot{\vec{R}}} - \left(\sum_i m_i \right) \dot{\vec{R}} \right] = \vec{R} \times [M\dot{\vec{R}} - M\dot{\vec{R}}] = \vec{0}. \quad (10)$$

Hence

$$\vec{L} = \vec{R} \times M\dot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i \equiv \vec{L}^{(0)}, \quad \frac{d\vec{L}^{(0)}}{dt} = \vec{R} \times M\ddot{\vec{R}} + \sum_i \vec{r}_i' \times m_i \ddot{\vec{r}}_i. \quad (11)$$

But also, directly,

$$\frac{d\vec{L}^{(0)}}{dt} = \vec{N}^{(e)} = \sum_i \vec{r}_i' \times \vec{F}_i^{(e)}, \quad \frac{d\vec{L}^{(0)}}{dt} = \vec{R} \times \sum_i \vec{F}_i^{(e)} + \sum_i \vec{r}_i' \times \vec{F}_i^{(e)} = \vec{R} \times M\ddot{\vec{R}} + \sum_i \vec{r}_i' \times \vec{F}_i^{(e)}, \quad (12)$$

using $M\ddot{\vec{R}} = \sum_i \vec{F}_i^{(e)}$ (Newton's second law for the particle system). Comparing Eqs. (11) and (12), the term $\vec{R} \times M\ddot{\vec{R}}$ cancels and

$$\frac{d\vec{L}'}{dt} = \sum_{i=1}^N \vec{r}_i' \times \vec{F}_i^{(e)}, \quad \vec{L}' \equiv \sum_i \vec{r}_i' \times m_i \dot{\vec{r}}_i. \quad (13)$$

This is the special case of Eq. (2) evaluated in the CM frame.

2 Angular Velocity as an Antisymmetric Rotation-Rate Matrix

For circular motion in a plane, $\vec{v} = \vec{\omega} \times \vec{r}$, with $\vec{v} \perp \vec{r}$ and $\vec{v} \cdot \vec{r} = 0$. Let $\hat{e}_i$ ($i = 1, 2, 3$) be an orthonormal, possibly time-dependent, basis:

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases} \quad (14)$$

For a rotating coordinate system, differentiating $\delta_{ij} = \hat{e}_i \cdot \hat{e}_j$,

$$0 = \frac{d\delta_{ij}}{dt} = \frac{d}{dt}(\hat{e}_i \cdot \hat{e}_j) = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}. \quad (15)$$

For $i = j$ this gives $0 = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt}$; for $i \neq j$ it gives $0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}$. Since $(\hat{e}_j)_{j=1,2,3}$ is a basis, $\frac{d\hat{e}_i}{dt}$ can be written as a linear combination of the $\hat{e}_j$:

$$\boxed{\frac{d\hat{e}_i}{dt} = \sum_{j=1}^3 \Omega_{ij} \hat{e}_j, \quad \Omega_{ij} \equiv \hat{e}_j \cdot \frac{d\hat{e}_i}{dt},} \quad (16)$$

using that $\hat{e}_j \cdot \sum_i \sigma_i \hat{e}_i = \sum_i \sigma_i \hat{e}_j \cdot \hat{e}_i = \sigma_j$. From Eq. (15), for $i \neq j$, $0 = \Omega_{ij} + \Omega_{ji} \Rightarrow \Omega_{ji} = -\Omega_{ij}$, and $\Omega_{ii} = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt} = 0$; that is, Ω_{ij} is antisymmetric,

$$\Omega = \begin{pmatrix} 0 & \Omega_{12} & -\Omega_{31} \\ -\Omega_{12} & 0 & \Omega_{23} \\ \Omega_{31} & -\Omega_{23} & 0 \end{pmatrix}. \quad (17)$$

An antisymmetric 3×3 matrix has three independent entries, matched to the components of an axial vector $\vec{\omega}$ by

$$\omega_1 \equiv \Omega_{23}, \quad \omega_2 \equiv \Omega_{31}, \quad \omega_3 \equiv \Omega_{12}. \quad (18)$$

Explicitly,

$$\frac{d\hat{e}_1}{dt} = \Omega_{12}\hat{e}_2 + \Omega_{13}\hat{e}_3 = \Omega_{12}\hat{e}_2 - \Omega_{31}\hat{e}_3 = \omega_3\hat{e}_2 - \omega_2\hat{e}_3 = \vec{\omega} \times \hat{e}_1, \quad (19)$$

$$\frac{d\hat{e}_2}{dt} = \Omega_{21}\hat{e}_1 + \Omega_{23}\hat{e}_3 = -\Omega_{12}\hat{e}_1 + \Omega_{23}\hat{e}_3 = -\omega_3\hat{e}_1 + \omega_1\hat{e}_3 = \vec{\omega} \times \hat{e}_2, \quad (20)$$

$$\frac{d\hat{e}_3}{dt} = \vec{\omega} \times \hat{e}_3, \quad (21)$$

where Eq. (19) is checked directly: $\vec{\omega} \times \hat{e}_1 = (\omega_1\hat{e}_1 + \omega_2\hat{e}_2 + \omega_3\hat{e}_3) \times \hat{e}_1 = -\omega_2\hat{e}_3 + \omega_3\hat{e}_2$, matching the left side; similarly for Eq. (20), using $\hat{e}_1 \times \hat{e}_2 = \hat{e}_3$ and $\hat{e}_3 \times \hat{e}_2 = -\hat{e}_1$.

2.1 The Levi-Civita Symbol and Cross Products

With $\hat{e}_1 \equiv \hat{x}$, $\hat{e}_2 \equiv \hat{y}$, $\hat{e}_3 \equiv \hat{z}$ (Fig. 4), so that $\hat{e}_1 \times \hat{e}_2 = \hat{e}_3$ (and cyclically), $\hat{e}_2 \times \hat{e}_3 = \hat{e}_1$, $\hat{e}_3 \times \hat{e}_1 = \hat{e}_2$, the Levi-Civita symbol is

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ a cyclic permutation of } 123 \\ -1 & ijk \text{ a cyclic permutation of } 213 \\ 0 & \text{if any index is repeated,} \end{cases} \quad (22)$$

so that $\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$, $\epsilon_{213} = \epsilon_{132} = \epsilon_{321} = -1$, $\epsilon_{iij} = \epsilon_{iji} = \epsilon_{jii} = 0$, and

$$\hat{e}_i \times \hat{e}_j = \sum_{\ell} \epsilon_{ij\ell} \hat{e}_{\ell}. \quad (23)$$

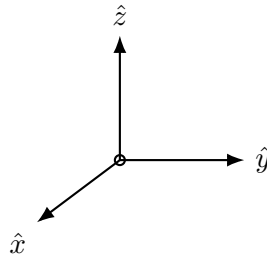


Figure 4: Right-handed orthonormal triad used for the Levi-Civita convention: $\hat{x} \times \hat{y} = \hat{z}$ and cyclically.

3 The Transport Theorem for Rotating Frames

Let $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ be a fixed inertial triad and $\hat{e}_1, \hat{e}_2, \hat{e}_3$ a rotating (non-inertial) triad (Fig. 5). For an arbitrary vector $\vec{b}(t)$ with components b_i^0 in the inertial basis and b_i in the rotating basis,

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i^0 \hat{e}_i^0 \right) = \sum_{i=1}^3 \dot{b}_i^0 \hat{e}_i^0, \quad \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} \equiv \sum_{i=1}^3 \frac{db_i}{dt} \hat{e}_i, \quad (24)$$

while resolving $\vec{b}$ in the rotating basis and differentiating in the inertial frame,

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \frac{d}{dt} \left(\sum_{i=1}^3 b_i \hat{e}_i \right) = \sum_{i=1}^3 \dot{b}_i \hat{e}_i + \sum_{i=1}^3 b_i \dot{\hat{e}}_i. \quad (25)$$

Using $\dot{\hat{e}}_i = \vec{\omega} \times \hat{e}_i$ (Eq. (19)),

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \sum_{i=1}^3 b_i (\vec{\omega} \times \hat{e}_i), \quad (26)$$

so that

$$\boxed{\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{b}.} \quad (27)$$

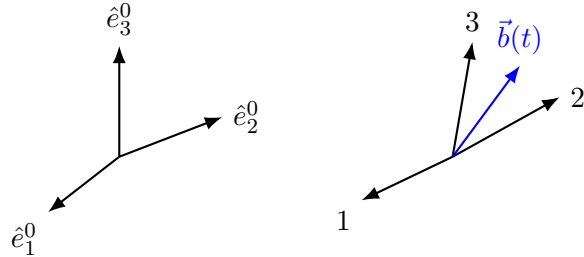


Figure 5: Fixed inertial triad $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ (left) and rotating body triad $\hat{e}_1, \hat{e}_2, \hat{e}_3$ (right, labeled 1, 2, 3), together with an arbitrary vector $\vec{b}(t)$ resolved in the rotating basis.

3.1 Rigid-Body Kinetic Energy from the Transport Theorem

For a rigid body with a fixed point, let $\vec{r}_P$ be the (constant, in the body frame) position of point P in the body frame. Since the body is rigid, $\vec{v}_P = \vec{\omega} \times \vec{r}_P$, and

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2. \quad (28)$$

Useful vector identities, verified using the Levi-Civita symbol:

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = \sum_{\ell} \epsilon_{12\ell} \hat{e}_{\ell}, \quad \vec{a} \times \vec{b} = \left(\sum_i a_i \hat{e}_i \right) \times \left(\sum_j b_j \hat{e}_j \right) = \sum_{ij\ell} a_i b_j \epsilon_{ij\ell} \hat{e}_{\ell}, \quad (29)$$

$$(\vec{a} \times \vec{b})_k = \sum_{ij} \epsilon_{ijk} a_i b_j, \quad (30)$$

and, using $\sum_i \epsilon_{jki} \epsilon_{lmi} = \delta_{j\ell} \delta_{km} - \delta_{jm} \delta_{k\ell}$ together with the cyclic property $\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}$,

$$\begin{aligned} (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= \sum_{ijk} A_i B_j \epsilon_{ijk} \hat{e}_k \cdot \sum_{lmn} C_\ell D_m \epsilon_{lmn} \hat{e}_n = \sum_{ijklm} \epsilon_{ijk} \epsilon_{lmk} A_i B_j C_\ell D_m \\ &= (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}). \end{aligned} \quad (31)$$

4 Rigid-Body Kinetic Energy and the Inertia Tensor

Applying Eq. (31) to Eq. (28) with $\vec{A} = \vec{C} = \vec{\omega}$, $\vec{B} = \vec{D} = \vec{r}_P$,

$$T = \sum_P \frac{m_P}{2} \left[\omega^2 r_P^2 - (\vec{\omega} \cdot \vec{r}_P)^2 \right]. \quad (32)$$

Writing this in components,

$$T = \frac{1}{2} \sum_P m_P \left[r_P^2 \sum_{i=1}^3 \omega_i \omega_i - \left(\sum_{i=1}^3 \omega_i r_{Pi} \right) \left(\sum_{j=1}^3 \omega_j r_{Pj} \right) \right] = \frac{1}{2} \sum_P m_P \sum_{ij=1}^3 \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right] \omega_i \omega_j, \quad (33)$$

so that

$$\boxed{T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j}, \quad \boxed{I_{ij} \equiv \sum_P m_P \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right]} \quad (\text{moment of inertia tensor, symmetric}). \quad (34)$$

Likewise for the angular momentum,

$$\vec{L} = \sum_P m_P \vec{r}_P \times \vec{v}_P = \sum_P m_P \vec{r}_P \times (\vec{\omega} \times \vec{r}_P). \quad (35)$$

Aside: Summation-Notation Identity

Using the epsilon-contraction identity of Eq. (31) in component form,

$$\begin{aligned} [\vec{A} \times (\vec{B} \times \vec{C})]_i &= \epsilon_{ijk} A_j \epsilon_{klm} B_\ell C_m = \epsilon_{kij} \epsilon_{klm} A_j B_\ell C_m \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j B_\ell C_m = (\vec{A} \cdot \vec{C}) B_i - (\vec{A} \cdot \vec{B}) C_i, \end{aligned} \quad (36)$$

using $\epsilon_{ijk} = \epsilon_{kij}$. Applying this to $\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)$ with $\vec{A} = \vec{B} = \vec{r}_P$, $\vec{C} = \vec{\omega}$,

$$[\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = (\vec{r}_P \cdot \vec{r}_P) \omega_i - (\vec{r}_P \cdot \vec{\omega}) r_{Pi}, \quad (37)$$

so that

$$L_i = \sum_P m_P [\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = \sum_P m_P \left[r_P^2 \omega_i - \sum_{j=1}^3 r_{Pi} r_{Pj} \omega_j \right] = \sum_{j=1}^3 \underbrace{\left(\sum_P m_P [r_P^2 \delta_{ij} - r_{Pi} r_{Pj}] \right)}_{=I_{ij}} \omega_j. \quad (38)$$

Collecting results,

$$\boxed{L_i = \sum_j I_{ij} \omega_j}, \quad \boxed{T = \frac{1}{2} \vec{\omega} \cdot \vec{L}}, \quad \boxed{T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j}. \quad (39)$$

5 Principal Axes

Since $I_{ij} = I_{ji} = I_{ji}^*$, the inertia tensor is Hermitian (here, real symmetric) and can therefore be diagonalized:

$$I_{ij} = I_{ji} = I_{ji}^* \Rightarrow \text{Hermitian} \Rightarrow \text{diagonalizable}, \quad \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}. \quad (40)$$

The directions that diagonalize I_{ij} are called the *principal axes*.

5.1 Examples

For a homogeneous cube or sphere, all three principal moments coincide, $I_{ij} = I\delta_{ij}$, i.e. $\text{diag}(I, I, I) = I\mathbf{1}_{3 \times 3}$. For an oblate body (radius b much larger than the half-thickness a , Fig. 6a), $I_1 = I_2 > I_3$; for a prolate body (half-length a much larger than the radius b , Fig. 6b), $I_1 = I_2 < I_3$. In general, for an irregular body, I_1, I_2, I_3 are all different.

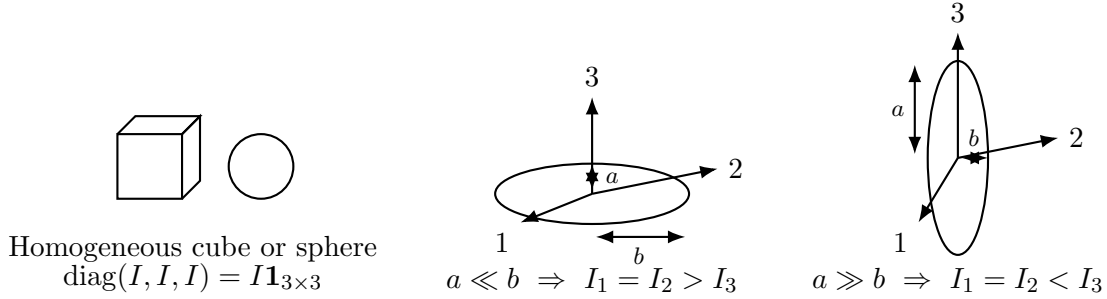


Figure 6: Principal-axis examples: (left) a homogeneous cube or sphere, for which all principal moments coincide; (center) an oblate disk, $a \ll b$; (right) a prolate rod, $a \gg b$.

These follow from $I = \sum_P m_P (\text{distance to axis})^2 = \sum_P m_P r_{P1}^2$ for rotation about axis 1: since $r_{P1}r_{P1} + r_{P2}r_{P2} + r_{P3}r_{P3} - r_{P3}r_{P3} = r_{P1}^2$, i.e. $x^2 + y^2 + z^2 - z^2 = x^2 + y^2 \equiv \rho^2$ for the axis 3 (using $I_{33} = \sum_P m_P [\delta_{33}r_P^2 - r_{P3}r_{P3}]$), one has $I_3 \sim Ma^2$, $I_1 \sim Mb^2$: if $a \ll b$ then $I_3 \ll I_1$, and if $a \gg b$ then $I_3 \gg I_1$.

5.2 Diagonal Form

In the principal-axis basis, $I_{ij} = I_i\delta_{ij}$, and

$$L_i = \sum_{j=1}^3 I_{ij}\omega_j = \sum_{j=1}^3 I_i\delta_{ij}\omega_j = I_i\omega_i, \quad \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} I_1\omega_1 \\ I_2\omega_2 \\ I_3\omega_3 \end{pmatrix}, \quad (41)$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_{i=1}^3 \omega_i L_i = \frac{1}{2} \sum_{i=1}^3 \omega_i I_i \omega_i = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2, \quad (42)$$

$$\boxed{L_i = I_i \omega_i}, \quad \boxed{T = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2}. \quad (43)$$

6 The Parallel-Axis Theorem (Tensor Form) and Motion about an Arbitrarily Moving Point

With $\bar{I}_{ij}$ the moment of inertia tensor with respect to the center of mass, the tensor parallel-axis theorem reads

$$I_{ij} = \bar{I}_{ij} + M[\delta_{ij}a^2 - a_i a_j], \quad \text{origin } O \longleftrightarrow \text{origin CM}, \quad (44)$$

where $\vec{a}$ is the (constant) vector from the center of mass to the arbitrary origin O ; the elementary version is $I = \bar{I} + Md^2$ (Fig. 7).

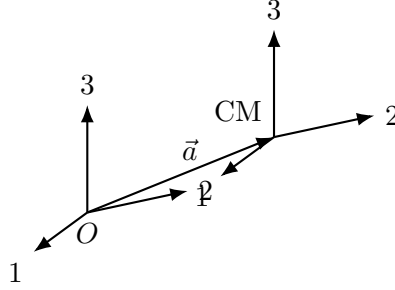


Figure 7: Two parallel triads, one at an arbitrary origin O and one at the center of mass, separated by the constant vector $\vec{a}$; $\bar{I}_{ij}$ is computed about the CM triad and I_{ij} about the triad at O .

Now consider rotation with respect to an arbitrarily moving point, i.e., no longer a point fixed in space. Three frames are in play: the inertial frame; the CM frame (origin at $\vec{R}$, axes parallel to the inertial axes, possibly accelerated but always parallel to the inertial frame — “translated but not rotated”); and the body frame (origin at $\vec{R}$, axes rotating with the body, so it is non-inertial even if $\ddot{\vec{R}} = \vec{0}$). Figure 8 shows the geometry: $\vec{r} = \vec{R} + \vec{r}'$, with $\vec{r}'$ the position relative to the center of mass.

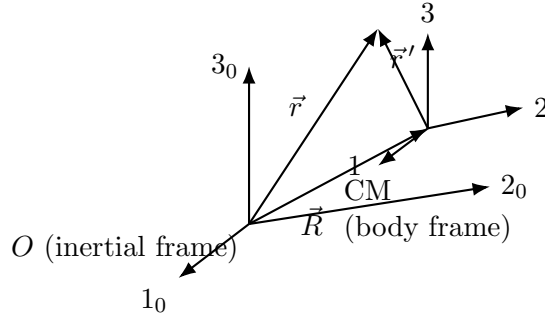


Figure 8: Position vector $\vec{r}$ from the inertial origin, resolved as $\vec{r} = \vec{R} + \vec{r}'$ with $\vec{R}$ the center-of-mass position and $\vec{r}'$ measured from the center of mass in the (rotating) body frame.

Applying the transport theorem, Eq. (27), to $\vec{r} = \vec{R} + \vec{r}'$,

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{inertial}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{CM}}, \quad \boxed{\left. \frac{d\vec{r}}{dt} \right|_{\text{inertial}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{r}'} \quad (45)$$

and, since $\vec{r}'$ is constant in the body frame for a rigid body,

$$\boxed{\text{if rigid body: } \left. \frac{d\vec{r}}{dt} \right|_{\text{inertial}} = \frac{d\vec{R}}{dt} + \vec{\omega} \times \vec{r}'} \quad (46)$$

For any system of particles, with $M = \sum_P m_P$ and $\vec{R} \equiv \sum_P m_P \vec{r}_P / \sum_P m_P$ the center-of-mass position,

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + T', \quad T' = \sum_P \frac{m_P}{2} \dot{\vec{r}}_P'^2, \quad \vec{L}^0 = \vec{R} \times M \dot{\vec{R}} + \vec{L}', \quad \vec{L}' = \sum_P \vec{r}_P' \times m_P \dot{\vec{r}}_P', \quad (47)$$

where T' and $\vec{L}'$ are measured in the frame with origin at the center of mass (the CM frame). For a rigid body, $\dot{\vec{r}}_P' = \vec{\omega} \times \vec{r}_P'$, so that

$$\vec{L}^0 = \vec{R} \times M \dot{\vec{R}} + \underbrace{\sum_P \vec{r}_P' \times m_P (\vec{\omega} \times \vec{r}_P')}_{\text{computed above}}, \quad T = \frac{1}{2} M \left| \dot{\vec{R}} \right|^2 + \underbrace{\sum_P \frac{m_P}{2} (\vec{\omega} \times \vec{r}_P')^2}_{\text{computed above}}, \quad (48)$$

with $L'_i = \sum_j \bar{I}_{ij} \omega_j$ and $T' = \frac{1}{2} \sum_{ij} \bar{I}_{ij} \omega_i \omega_j$, exactly as in Eqs. (34)–(39) but with the inertia tensor now understood as the tensor with respect to the center of mass, $\bar{I}_{ij}$.

7 Euler's Equations

Working in the CM frame ($\vec{R}$ possibly accelerated but always parallel to the inertial frame — translated but not rotated), Eq. (2) together with the transport theorem, Eq. (27), gives

$$\vec{N} = \frac{d\vec{L}}{dt} = \frac{d\vec{L}}{dt} \Big|_{\text{body}} + \vec{\omega} \times \vec{L}, \quad (49)$$

so that, component by component,

$$N_1 = \frac{dL_1}{dt} \Big|_{\text{body}} + (\vec{\omega} \times \vec{L})_1 = \frac{dL_1}{dt} \Big|_{\text{body}} + \omega_2 L_3 - \omega_3 L_2. \quad (50)$$

In the principal-axis basis $L_i = I_i \omega_i$, and since a rigid body has $\dot{\vec{r}}_P = \vec{0}$ in the body frame, $\frac{dL_i}{dt} \Big|_{\text{body}} = 0$, so

$$N_1 = I_1 \frac{d\omega_1}{dt} \Big|_{\text{body}} + \omega_2 I_3 \omega_3 - \omega_3 I_2 \omega_2 = I_1 \frac{d\omega_1}{dt} \Big|_{\text{body}} + \omega_2 \omega_3 (I_3 - I_2), \quad (51)$$

and cyclically. Euler's equations are

$$\boxed{\begin{aligned} I_1 \frac{d\omega_1}{dt} \Big|_{\text{body}} &= \omega_2 \omega_3 (I_2 - I_3) + N_1, \\ I_2 \frac{d\omega_2}{dt} \Big|_{\text{body}} &= \omega_3 \omega_1 (I_3 - I_1) + N_2, \\ I_3 \frac{d\omega_3}{dt} \Big|_{\text{body}} &= \omega_1 \omega_2 (I_1 - I_2) + N_3. \end{aligned}} \quad (52)$$

8 Application: Kater's Reversible Pendulum

Consider a rigid pendulum with one degree of freedom, so that a single angle Φ serves as the generalized coordinate, free of friction, with body axes $\hat{e}_1, \hat{e}_2, \hat{e}_3$ fixed at the pivot (Fig. 9). In the lab frame, $\omega_3 = \dot{\Phi}$, and

$$I\dot{\omega}_3 = \frac{dL_3}{dt} = N_3 = -Mgl \sin \Phi, \quad (53)$$

since, with $\vec{g} = g(\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2)$,

$$\vec{N}_g = l\hat{e}_1 \times M\vec{g} = l\hat{e}_1 \times Mg(\cos \Phi \hat{e}_1 - \sin \Phi \hat{e}_2) = -Mgl \sin \Phi \hat{e}_3, \quad (54)$$

using $\hat{e}_1 \times \hat{e}_1 = 0$ and $\hat{e}_1 \times \hat{e}_2 = \hat{e}_3$. For small angles, $\sin \Phi \approx \Phi$, and

$$I\ddot{\Phi} \approx -Mgl\Phi, \quad \ddot{\Phi} \approx -\frac{Mgl}{I}\Phi, \quad \ddot{\Phi} = -\Omega^2\Phi, \quad \Omega = \left(\frac{Mgl}{I}\right)^{1/2} \quad (\text{frequency of oscillation}). \quad (55)$$

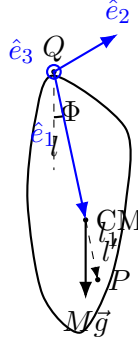


Figure 9: Kater's reversible pendulum: body axes $\hat{e}_1, \hat{e}_2, \hat{e}_3$ fixed at the pivot Q , center of mass a distance l from Q , percussion center P a distance l_1 from Q along the line through the center of mass, and generalized coordinate Φ measured from the vertical.

8.1 Radius of Gyration and Reversibility

Define the radius of gyration $\bar{k}$ about the center of mass by $\bar{I} \equiv M\bar{k}^2$, and the radius of gyration k about Q by $I \equiv Mk^2$; the parallel-axis theorem $I = \bar{I} + Ml^2$ then gives $k^2 = \bar{k}^2 + l^2$. Then

$$\Omega_Q^2 = \frac{Mgl}{M(\bar{k}^2 + l^2)} = \frac{g}{l_1}, \quad l_1 \equiv \frac{k^2}{l} = l + \frac{\bar{k}^2}{l}, \quad (56)$$

with Ω_Q having units of inverse time, as required, since g/l_1 has units of acceleration over length. Comparing with the simple pendulum (point mass, length l), for which $\Omega = \sqrt{g/l}$: since a point mass has $\bar{I} = 0 \Rightarrow \bar{k} = 0$, indeed $l_1 = l + \bar{k}^2/l = l$, recovering the simple-pendulum result.

More generally,

$$\bar{I} = \int d^3r \rho(\vec{r}) (r_1^2 + r_2^2), \quad \bar{k}^2 = \frac{\bar{I}}{M} = \frac{\int d^3r \rho(\vec{r}) (r_1^2 + r_2^2)}{\int d^3r \rho(\vec{r})}, \quad (57)$$

i.e., the mean-square distance to the rotation axis weighted by the mass density $\rho(\vec{r})$, with r_1, r_2 measured relative to the center of mass. Suspend the pendulum instead from the percussion center

P , at distance l_1 from Q along the direction through the center of mass; then

$$\Omega_P^2 = \frac{gl'}{\bar{k}^2 + l'^2} = \frac{g(l_1 - l)}{\bar{k}^2 + (l_1 - l)^2} = \frac{g\bar{k}^2/l}{\bar{k}^2 + \bar{k}^4/l^2} = \frac{gl}{l^2 + \bar{k}^2} = \Omega_Q^2, \quad (58)$$

using $l_1 = l + \bar{k}^2/l \Rightarrow l_1 - l = \bar{k}^2/l \Rightarrow (l_1 - l)^2 = \bar{k}^4/l^2$. The oscillation frequency is thus identical whether the pendulum is suspended from Q or from the percussion center P a distance l_1 away — the basis of Kater's method for measuring g .

9 Torque-Free Motion of a Symmetric Top

For torque-free motion of a symmetric top, $I_1 = I_2 \neq I_3$, Euler's equations (52) with $N_i = 0$ reduce to

$$I_1 \frac{d\omega_1}{dt} \Big|_{\text{body}} = \omega_2 \omega_3 (I_2 - I_3) \Rightarrow \dot{\omega}_1 = - \left[\omega_3 \frac{I_3 - I_2}{I_1} \right] \omega_2, \quad (59)$$

$$I_2 \frac{d\omega_2}{dt} \Big|_{\text{body}} = \omega_3 \omega_1 (I_3 - I_1) \Rightarrow \dot{\omega}_2 = \left[\omega_3 \frac{I_3 - I_1}{I_2} \right] \omega_1, \quad (60)$$

and $I_3 \dot{\omega}_3 = \omega_1 \omega_2 (I_1 - I_2) = 0$. Defining the precession rate

$$\Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1}, \quad (61)$$

and using $I_1 = I_2$,

$$\boxed{\dot{\omega}_1 = -\Omega \omega_2}, \quad \boxed{\dot{\omega}_2 = \Omega \omega_1}, \quad \boxed{\dot{\omega}_3 = 0}. \quad (62)$$

9.1 Precession

Equations (62) are equivalent to

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} \iff \vec{\omega} \text{ rotates with angular velocity } \Omega \hat{e}_3 \Rightarrow \text{precession}, \quad (63)$$

since

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} = \Omega \hat{e}_3 \times (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) = \Omega(\omega_1 \hat{e}_2 - \omega_2 \hat{e}_1), \quad (64)$$

which reproduces $\dot{\omega}_1 = -\Omega \omega_2$, $\dot{\omega}_2 = \Omega \omega_1$, $\dot{\omega}_3 = 0$: $\vec{\omega}$ rotates about $\hat{e}_3$ with angular speed Ω , sweeping out the precession cone of Fig. 10. If initially $\vec{\omega} = \omega(\sin \lambda \hat{e}_1 + \cos \lambda \hat{e}_3)$, so that $\vec{\omega}$ lies in the plane of $\hat{e}_1$ and $\hat{e}_3$, then

$$\boxed{\omega_1 = \omega \sin \lambda \cos(\Omega t)}, \quad \boxed{\omega_2 = \omega \sin \lambda \sin(\Omega t)}, \quad \boxed{\omega_3 = \omega \cos \lambda}. \quad (65)$$

10 Euler Angles

Let $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ be the inertial set of unit vectors (not rotating) and $\hat{e}_1, \hat{e}_2, \hat{e}_3$ the body set of unit vectors (rotating with the body). Six coordinates are needed to describe the positions of all points in a rigid body: the center-of-mass position $\vec{R} \leftrightarrow (X, Y, Z)$, and three coordinates to fix the orientation of $\hat{e}_1, \hat{e}_2, \hat{e}_3$ with respect to $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$. These orientation coordinates are chosen to be the Euler angles α, β, γ :

- $\hat{e}_\beta \perp \hat{e}_3^0, \hat{e}_3$;

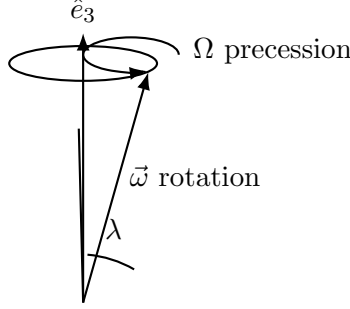


Figure 10: Torque-free precession of a symmetric top: the angular velocity $\vec{\omega}$ makes a constant angle λ with the body symmetry axis $\hat{e}_3$ and sweeps out a cone at the precession rate Ω .

- β is the polar angle and α the azimuthal angle orienting the new “north pole” $\hat{e}_3$ relative to $\hat{e}_3^0$;
- γ specifies the additional rotation about $\hat{e}_3$;
- α, β, γ can be chosen independently and completely specify the orientation of the body frame $\Rightarrow$ use α, β, γ as generalized coordinates.

Following Chapter 5 of Fetter and Walecka (skipping a considerable amount of intervening development), the kinetic energy and angular velocity are

$$T = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2), \quad \vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\beta} \hat{e}_\beta + \dot{\gamma} \hat{e}_3, \quad \hat{e}_\beta = \cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1, \quad (66)$$

$$\hat{e}_3^0 = \cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2], \quad (67)$$

$$\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3. \quad (68)$$

10.1 Infinitesimal Rotations Add

For infinitesimal rotations, $R_\alpha = \mathbf{1} + M_\alpha \dot{\alpha} dt$ and $R_\beta = \mathbf{1} + M_\beta \dot{\beta} dt$, so that, to order dt ,

$$R_\alpha R_\beta = \mathbf{1} + (M_\alpha \dot{\alpha} + M_\beta \dot{\beta}) dt + \mathcal{O}(dt^2), \quad R_\beta R_\alpha = \mathbf{1} + (M_\beta \dot{\beta} + M_\alpha \dot{\alpha}) dt + \mathcal{O}(dt^2), \quad (69)$$

which are equal to order dt ; hence infinitesimal rotations add,

$$\vec{\omega} = \vec{\omega}_\alpha + \vec{\omega}_\beta + \vec{\omega}_\gamma, \quad (70)$$

consistent with Eq. (66).

11 Kinetic Energy in Euler Angles; the Symmetric-Top Lagrangian

Substituting Eq. (68) into $T = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2)$,

$$T = \frac{I_1}{2} (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)^2 + \frac{I_2}{2} (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)^2 + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2. \quad (71)$$

For a symmetric top, $I_1 = I_2$, this simplifies to

$$T = \frac{I_1}{2} (\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2, \quad (72)$$

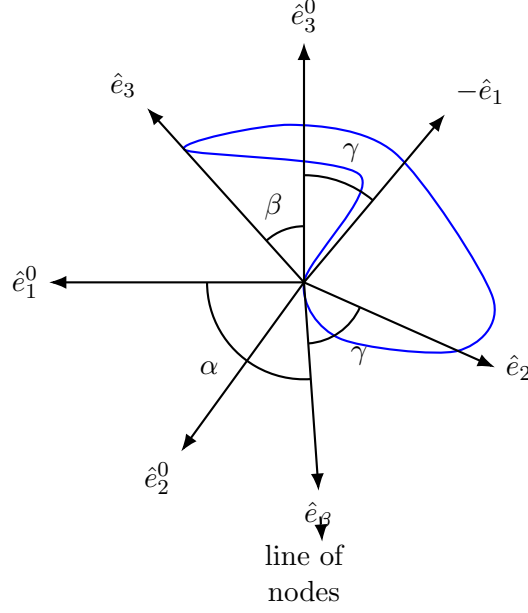


Figure 11: The Euler angles α (azimuthal), β (polar), γ (spin about $\hat{e}_3$) relating the inertial triad $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$ to the body triad $\hat{e}_1, \hat{e}_2, \hat{e}_3$. The line of nodes is the intersection of the $\hat{e}_1^0$ - $\hat{e}_2^0$ and $\hat{e}_1$ - $\hat{e}_2$ planes and carries the unit vector $\hat{e}_\beta$; the blue curve traces the (edge-on) body equatorial plane spanned by $\hat{e}_1, \hat{e}_2$.

and, since there is no potential energy contribution to the orientation coordinates for torque-free motion, the Lagrangian is

$$L = T = \frac{I_1}{2}(\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2}(\dot{\gamma} + \dot{\alpha} \cos \beta)^2. \quad (73)$$

With generalized coordinates α, β, γ , there are three Lagrange equations; α and γ are cyclic (absent from L), giving two conserved momenta:

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta, \quad p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3(\dot{\gamma} + \dot{\alpha} \cos \beta). \quad (74)$$

It will be shown below that $p_\alpha = \vec{L} \cdot \hat{e}_3^0$ and $p_\gamma = \vec{L} \cdot \hat{e}_3$.

11.1 Verifying $p_\gamma = \vec{L} \cdot \hat{e}_3$

Since $p_\gamma = I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) = I_3\omega_3$ by Eq. (68), and, using

$$\vec{L} = I_1(\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3\omega_3 \hat{e}_3, \quad (75)$$

one has immediately $\vec{L} \cdot \hat{e}_3 = I_3\omega_3 = p_\gamma$, as anticipated.

11.2 Verifying $p_\alpha = \vec{L} \cdot \hat{e}_3^0$

Compare $p_\alpha = I_1 \sin^2 \beta \dot{\alpha} + I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta$ with

$$\vec{L} \cdot \hat{e}_3^0 = \left(I_1(\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3\omega_3 \hat{e}_3 \right) \cdot \left(\cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \right), \quad (76)$$

using Eq. (67). Expanding,

$$\begin{aligned}\vec{L} \cdot \hat{e}_3^0 &= I_1 \left[(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)(-\sin \beta \cos \gamma) \right. \\ &\quad \left. + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)(\sin \beta \sin \gamma) \right] + I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \\ &= I_1 \dot{\alpha} \sin^2 \beta (\cos^2 \gamma + \sin^2 \gamma) + I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta,\end{aligned}\tag{77}$$

so that

$$\boxed{\vec{L} \cdot \hat{e}_3^0 = I_1 \dot{\alpha} \sin^2 \beta + I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta = p_\alpha}, \quad \checkmark \tag{78}$$

matching Eq. (74).

11.3 The Momentum Conjugate to β

Since β appears explicitly in the Lagrangian, $p_\beta \equiv \frac{\partial L}{\partial \dot{\beta}} = I_1 \dot{\beta}$ is not, in principle, a constant of the motion. Its time derivative follows from

$$\frac{\partial L}{\partial \beta} = I_1 \sin \beta \cos \beta \dot{\alpha}^2 - I_3(\dot{\gamma} + \dot{\alpha} \cos \beta) \dot{\alpha} \sin \beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3), \tag{79}$$

so that, by the Euler-Lagrange equation,

$$\dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (\text{however, see below}). \tag{80}$$

Comparing $p_\beta = I_1 \dot{\beta}$ with

$$\begin{aligned}\vec{L} \cdot \hat{e}_\beta &= \left(I_1(\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \right) \cdot (\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1) \\ &= I_1 \left[(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \sin \gamma + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \cos \gamma \right],\end{aligned}\tag{81}$$

using Eq. (66), gives

$$\vec{L} \cdot \hat{e}_\beta = I_1 \dot{\beta} (\sin^2 \gamma + \cos^2 \gamma) = I_1 \dot{\beta} \Rightarrow \boxed{\vec{L} \cdot \hat{e}_\beta = p_\beta}. \quad \checkmark \tag{82}$$

12 Description in the Inertial (or CM) Frame

Since there is no torque, $\dot{\vec{L}} = \vec{0}$. Choose the inertial $\hat{e}_3^0$ axis along the (fixed) direction of $\vec{L}$, so that $\vec{L} \cdot \hat{e}_3^0 = |\vec{L}|$ and $\vec{L} = |\vec{L}| \hat{e}_3^0$; the magnitude $|\vec{L}|$ does not change. Since $\vec{L} \cdot \hat{e}_3 = p_\gamma$ is a constant of the motion,

$$\vec{L} \cdot \hat{e}_3 = |\vec{L}| \hat{e}_3^0 \cdot \hat{e}_3 \Rightarrow \hat{e}_3^0 \cdot \hat{e}_3 = \frac{p_\gamma}{|\vec{L}|} = \text{const}, \tag{83}$$

and since $\hat{e}_3^0 \cdot \hat{e}_3 = \cos(\text{angle between } \hat{e}_3 \text{ and } \hat{e}_3^0) = \cos \beta$, β is a constant of the motion:

$$\beta = \text{const} \Rightarrow \dot{\beta} = 0 \Rightarrow p_\beta = I_1 \dot{\beta} = \vec{L} \cdot \hat{e}_\beta = 0, \tag{84}$$

i.e., for this particular choice of axes, both p_β and $\vec{L} \cdot \hat{e}_\beta$ vanish. Since $\vec{L} \cdot \hat{e}_\beta = 0$ and $\vec{L} \neq \vec{0}$, $|\hat{e}_\beta| = 1 \neq 0$, so

$$\vec{L} \perp \hat{e}_\beta \Rightarrow \vec{L} \text{ lies in the plane of } \hat{e}_3 \text{ and } \hat{e}_3^0. \tag{85}$$

The constants of the motion are $I_1, I_3, \beta, I_3\omega_3 = L_3, p_\alpha$. From Eq. (74),

$$p_\alpha = I_1 \sin^2 \beta \dot{\alpha} + I_3 \omega_3 \cos \beta = \text{const} \Rightarrow \dot{\alpha} = \text{const}, \quad (86)$$

since $I_1, \beta, I_3\omega_3$ are all constant. Likewise, from Eq. (74), since $I_3, p_\gamma, \dot{\alpha}, \beta$ are constant, $\dot{\gamma} = \text{const}$. Hence

$$\alpha = \dot{\alpha}_0 t, \quad \gamma = \dot{\gamma}_0 t, \quad \dot{\alpha}_0, \dot{\gamma}_0 \text{ constants.} \quad (87)$$

12.1 Relating $\dot{\gamma}$ to the Precession Rate

Setting $\dot{p}_\beta = 0$ in Eq. (80),

$$0 = \dot{p}_\beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \Rightarrow \dot{\alpha} \cos \beta = \frac{I_3 \omega_3}{I_1} \quad (88)$$

(for $\sin \beta \neq 0, \dot{\alpha} \neq 0$). Then, from Eq. (68), $\omega_3 = \dot{\gamma} + \dot{\alpha} \cos \beta = \dot{\gamma} + I_3 \omega_3 / I_1$, so

$$\boxed{\dot{\gamma} = \omega_3 \frac{I_1 - I_3}{I_1} = -\Omega,} \quad (89)$$

with $\Omega \equiv \omega_3(I_3 - I_1)/I_1$ the precession rate of Eq. (61). With $\dot{\beta} = 0$, Eq. (66) reduces to

$$\vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\gamma} \hat{e}_3 \Rightarrow \hat{e}_3, \hat{e}_3^0, \vec{\omega}, \vec{L} \text{ all lie in one plane.} \quad (90)$$

Consider the two cases $I_1 = I_2 < I_3$ and $I_1 = I_2 > I_3$, with $\vec{L} = I_1(\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3$ as in Eq. (75).

Closing Remark

The simple-pendulum equation of motion, $\ddot{\theta} = -\frac{g}{l} \sin \theta$, has exactly the same functional form as the equation governing the corresponding tilt angle of a symmetric top pivoted at a point (the “thumbtack” problem), $(\text{angle})'' = -C \sin(\text{angle})$, except that the coefficient C is now an unwieldy combination of the principal moments of inertia and the precession rate Ω , rather than the simple ratio g/l .